

Multiplying by a One-Digit Number

Answer Key

Grades 4–5 Number and Operations in Base Ten

Quick Answers

1. 642	4. 3729	7. 7650	10. 9030
2. 528	5. 24000, 25122	8. 21992	11. 44775
3. 2380	6. 18040	9. 40600, 40684	12. 14800

Curriculum

Level: Grades 4–5 Number and Operations in Base Ten

4.NBT.B.5 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*

Difficulty 1–5 of 5 across 15 tagged checks.

Common wrong answers

6. *If they got 1840: skipped the zero in the tens place and multiplied 368 by 5*
 10. *If they got 9000: wrote 0 in the tens place and never added the regrouped 3 to it*
 12. *If they got 18800: found the total ordered but forgot to subtract the sheets already used*
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1. Multiply 214×3 .

Step 1. Multiply the ones: $4 \times 3 = 12$. Write the 2 in the ones place and regroup the 1 into the tens column.

Step 2. Multiply the tens: $1 \times 3 = 3$, then add the regrouped 1 to get 4 tens.

Step 3. Multiply the hundreds: $2 \times 3 = 6$ hundreds. Nothing to regroup.

642

Listen for: a student who adds the regrouped digit before multiplying. Multiply first, add the carry second.

2. Multiply 132×4 .

Step 1. Ones: $2 \times 4 = 8$. Write 8; there is nothing to regroup.

Step 2. Tens: $3 \times 4 = 12$. Write the 2 in the tens place and regroup the 1 into the hundreds column.

Step 3. Hundreds: $1 \times 4 = 4$, plus the regrouped 1 makes 5 hundreds.

528

Teaching note: the regroup out of the tens is the one students most often drop, because the ones column did not need one.

3. Multiply 476×5 .

Step 1. Ones: $6 \times 5 = 30$. Write 0, regroup 3.

Step 2. Tens: $7 \times 5 = 35$, plus the regrouped 3 is 38. Write 8 in the tens place, regroup 3.

Step 3. Hundreds: $4 \times 5 = 20$, plus the regrouped 3 is 23. That is 23 hundreds, so both digits are written and the product grows to four digits. 2380

Teaching note: a three-digit number times five is often a four-digit product — an answer with only three digits is worth a second look.

4. Multiply 1243×3 .

Step 1. Ones: $3 \times 3 = 9$. Tens: $4 \times 3 = 12$, so write 2 and regroup 1.

Step 2. Hundreds: $2 \times 3 = 6$, plus the regrouped 1 is 7.

Step 3. Thousands: $1 \times 3 = 3$. Reading the columns back together gives the product. 3729

Check it: the factor is a bit over 1200 and $1200 \times 3 = 3600$, so an answer near 3700 is sensible.

5. Estimate 4187×6 by rounding, then find the exact product.

Step 1. 4187 is between 4000 and 5000, and the hundreds digit 1 is less than 5, so to the nearest thousand it rounds down to 4000.

Step 2. Estimate with the rounded factor: 4000×6 is $4 \times 6 = 24$ thousands. 24000

Step 3. Exact product. Ones: $7 \times 6 = 42$, write 2 regroup 4. Tens: $8 \times 6 = 48$, plus 4 is 52 — write 2, regroup 5. Hundreds: $1 \times 6 = 6$, plus 5 is 11 — write 1, regroup 1. Thousands: $4 \times 6 = 24$, plus 1 is 25. 25122

Compare: the estimate is low by about 1100, which is what rounding a factor down by 187 costs.

6. Multiply 3608×5 .

Step 1. Ones: $8 \times 5 = 40$. Write 0, regroup 4.

Step 2. Tens: the digit is 0, and $0 \times 5 = 0$ — but the regrouped 4 still has to be added, so the tens digit is 4.

Step 3. Hundreds: $6 \times 5 = 30$. Write 0, regroup 3. Thousands: $3 \times 5 = 15$, plus 3 is 18. 18040

Common wrong answer: 1840 — the zero column was skipped altogether and 368×5 was computed instead. A zero digit still holds a place.

7. A bakery loads 1275 rolls into each of 6 vans.

Step 1. Six vans each holding the same number of rolls is six equal groups, so the total is 1275×6 .

Step 2. Ones: $5 \times 6 = 30$, write 0 regroup 3. Tens: $7 \times 6 = 42$, plus 3 is 45 — write 5, regroup 4.

Step 3. Hundreds: $2 \times 6 = 12$, plus 4 is 16 — write 6, regroup 1. Thousands: $1 \times 6 = 6$, plus 1 is 7. 7650

Check it: 1275 is close to 1300, and $1300 \times 6 = 7800$, so a total near 7650 is reasonable.

8. Multiply 2749×8 .

Step 1. Ones: $9 \times 8 = 72$. Write 2, regroup 7.

Step 2. Tens: $4 \times 8 = 32$, plus 7 is 39 — write 9, regroup 3. Hundreds: $7 \times 8 = 56$, plus 3 is 59 — write 9, regroup 5.

Step 3. Thousands: $2 \times 8 = 16$, plus 5 is 21, and both digits are written.

21992

Listen for: regroupings larger than one digit surprise students here. The carry can be as big as 8; it is still a single digit.

9. Estimate 5812×7 by rounding, then find the exact product.

Step 1. To the nearest hundred, 5812 rounds to 5800: the tens digit 1 is less than 5, so the hundreds digit stays put.

Step 2. Estimate: 5800×7 is $58 \times 7 = 406$ hundreds.

40600

Step 3. Exact product. Ones: $2 \times 7 = 14$, write 4 regroup 1. Tens: $1 \times 7 = 7$, plus 1 is 8. Hundreds: $8 \times 7 = 56$, write 6 regroup 5. Thousands: $5 \times 7 = 35$, plus 5 is 40.

40684

Compare: rounding to the nearest hundred left the estimate only 84 away, while rounding to the nearest thousand would have missed by 1284.

10. Dana wrote $1806 \times 5 = 9000$. Find the mistake and the correct product.

Step 1. Ones: $6 \times 5 = 30$, so the ones digit is 0 and 3 is regrouped into the tens column.

Step 2. Tens: the digit is 0 and $0 \times 5 = 0$, but the regrouped 3 must still be added, so the tens digit is 3, not 0. This is Dana's mistake: she wrote the zero and dropped the carry.

Step 3. Hundreds: $8 \times 5 = 40$ — write 0, regroup 4. Thousands: $1 \times 5 = 5$, plus 4 is 9.

9030

Open response (manual review): the written explanation is the student's own. Accept any answer that says the regrouped 3 from the ones column was never added into the tens column.

Common wrong answer: 9000 — exactly Dana's slip.

11. Multiply 4975×9 .

Step 1. Ones: $5 \times 9 = 45$. Write 5, regroup 4.

Step 2. Tens: $7 \times 9 = 63$, plus 4 is 67 — write 7, regroup 6. Hundreds: $9 \times 9 = 81$, plus 6 is 87 — write 7, regroup 8.

Step 3. Thousands: $4 \times 9 = 36$, plus 8 is 44.

44775

Check it: 4975 is just under 5000 and $5000 \times 9 = 45000$, so a product a little below 45000 is what we expect.

12. 8 cartons of 2350 sheets, with 4000 sheets used.

Step 1. Eight equal cartons make this a multiplication first: the order holds 2350×8 sheets. The sheets already used are then taken away, so the question is a two-step problem.

Step 2. Ones: $0 \times 8 = 0$. Tens: $5 \times 8 = 40$ — write 0, regroup 4. Hundreds: $3 \times 8 = 24$, plus 4 is 28 — write 8, regroup 2. Thousands: $2 \times 8 = 16$, plus 2 is 18. The order is 18800 sheets.

Step 3. Subtract the sheets used: $18800 - 4000 = 14800$.

14800

Common wrong answer: 18800 — the multiplication was done and the subtraction forgotten. Re-reading the final sentence catches it.